

Auteur: Seda Jusjajeva
 Studierichting: Informatica
 Oefeningenreeks 5M nr. 2

$$\begin{cases} x(t) = a \cos^3 t \\ y(t) = a \sin^3 t \end{cases}$$

5M-2-seda.jusjajeva.INF.png

$$\begin{aligned} S &= 2 \cdot 2\pi \int_0^{\frac{\pi}{2}} |a \sin^3 t| \sqrt{((a \cos^3 t)')^2 + ((a \sin^3 t)')^2} dt \\ &= \int |a \sin^3 t| \sqrt{((a \cos^3 t)')^2 + ((a \sin^3 t)')^2} dt \\ &= a \int \sin^3 t \sqrt{(-3a \cos^2 t \sin t)^2 + (3a \sin^2 t \cos t)^2} dt \\ &= a \int \sin^3 t \sqrt{9a^2 \cos^4 t \sin^2 t + 9a^2 \sin^4 t \cos^2 t} dt \\ &= 3a^2 \int \sin^3 t \sqrt{\cos^4 t \sin^2 t + \sin^4 t \cos^2 t} dt \\ &= 3a^2 \int \sin^3 t \sqrt{\cos^2 t \sin^2 t (\cos^2 t + \sin^2 t)} dt \\ &= 3a^2 \int \sin^3 t \cos t \sin t dt \\ &= 3a^2 \int \sin^4 t \cos t dt \end{aligned}$$

Substitutie: $s = \sin t \Rightarrow ds = \cos t dt$

$$= 3a^2 \int s^4 ds$$

$$= \frac{3a^2 s^5}{5} + c = \frac{3a^2 \sin^5 t}{5} + c$$

$$\Rightarrow S = 4\pi \left[\frac{3a^2 \sin^5 t}{5} \right]_0^{\frac{\pi}{2}} = \frac{12a^2 \pi \sin^5 \frac{\pi}{2}}{5} = \frac{12a^2 \pi}{5}$$

References